
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Mwesigye, Peter
Date of birth:	9/19/1978
Subject:	080-Principles of Flight
Result:	42,5 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	187	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
2	213	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
3	148	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
4	231	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
5	241	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
6	216	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
7	226	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
8	201	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
9	146	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
10	215	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
11	199	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
12	184	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
13	194	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
14	210	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
15	189	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
16	130	81. PRINCIPLES OF FLIGHT AEROPLANE		0,00	1,00
17	149	81. PRINCIPLES OF FLIGHT AEROPLANE		1,00	1,00
18	82	81.1.1.1 Laws and definitions		1,00	1,00
19	55	81.1.2.4 Centre of pressure and aerodynamic centre		0,00	1,00
20	84	81.1.3.1 The lift coefficient C_l		1,00	1,00
21	2	81.1.3.2 The drag coefficient C_d		1,00	1,00
22	93	81.1.4.2 Induced drag		0,00	1,00
23	74	81.1.4.2 Induced drag		0,00	1,00
24	57	81.1.4.2 Induced drag		0,00	1,00
25	11	81.1.4.2 Induced drag		0,00	1,00
26	35	81.1.6 Ground effect		0,00	1,00
27	56	81.1.6 Ground effect		0,00	1,00
28	38	81.1.8.2 The stall speed		1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	78	81.1.12.1	Ice and other contaminants	1,00	1,00
30	52	81.2.2.1	Normal shock waves	0,00	1,00
31	41	81.4.1.1	Basics and definitions	0,00	1,00
32	15	81.4.4.4	Factors affecting static directional stability	0,00	1,00
33	98	81.5.8.3	Stabiliser trim	0,00	1,00
34	77	81.5.8.3	Stabiliser trim	1,00	1,00
35	73	81.7.1.4	Propeller efficiency versus speed	0,00	1,00
36	8	81.7.4.2	Gyroscopic precession	0,00	1,00
37	33	81.7.4.2	Gyroscopic precession	0,00	1,00
38	1	81.7.4.4	Asymmetric blade effect	0,00	1,00
39	89	81.8.1.3	Straight steady descent	0,00	1,00
40	31	81.8.1.4	Straight steady glide	1,00	1,00
Total:42%				17,00	40,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Mwesigye, Peter

Location: Uganda Civil Aviation Authority

Your Result: 17,00 from 40,00 Points (42,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	17,00	40,00	42,50
81.1. SUBSONIC AERODYNAMICS	12	5,00	12,00	41,67
81.2. HIGH SPEED AERODYNAMICS	1	0,00	1,00	0,00
81.4. STABILITY	2	0,00	2,00	0,00
81.5. CONTROL	2	1,00	2,00	50,00
81.7. PROPELLERS	4	0,00	4,00	0,00
81.8. FLIGHT MECHANICS	2	1,00	2,00	50,00
Total	40	17,00	40,00	42,50

1

Which characteristic of a spin is not a characteristic of a steep spiral?

☐ High rate of rotation.

☐ Rapid loss of altitude.

☒ Stalled wing.

2

Which is true relating to the direction in which turns should be made during slope soaring?

- ☒ All turns should be made downwind toward the slope.
- ☐ All reversing turns should be made into the wind away from the slope.
- ☐ All reversing turns should be made to the left.

3

When a glider is turning in flight, the force that opposes the inward turning force is called

- ☐ adverse yaw.
- ☒ centrifugal force.
- ☐ resultant force.

4

If both the ram air input and drain hole of the pitot system are blocked, what airspeed indication can be expected?

- ☒ **Constant indicated airspeed during a descent.**
- ☐ **Decrease of indicated airspeed during a climb.**
- ☐ **No variation of indicated airspeed in level flight even if large power changes are made.**

5

(Refer to Figure 147.) Which is the correct sequence for recovery from the unusual attitude indicated?

See Attachments:

1. instrument_147.jpg

figure 147

(Attachment
No.1)

- ☐ Stop turn by raising right wing and add power at the same time, lower the nose, and return to original attitude and heading.
- ☒ Add power, lower nose, level wings, return to original attitude and heading.
- ☐ Level wings, add power, lower nose, descend to original attitude, and heading.

6

It may be possible to make changes in the direction of flight of a hot air balloon by

- ☒ using the maneuvering vent.
- ☐ flying a constant atmospheric pressure gradient.
- ☐ operating at different flight altitudes.

7

During a wing stall, the wing tips of a weight shift aircraft are

- ☐ ineffective for stall recovery.
- ☒ effective only when combined with maximum engine output.
- ☐ effective for stall recovery.

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Subject: 080-Principles of Flight
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8

A rapid acceleration can create the illusion of being in a

☐ left turn.

☐ nosedown attitude.

☒ noseup attitude.

9

Which statement is generally true regarding wing camber of a glider's airfoil?

- ☐ The camber is the same on both the upper and lower surface of the wing.
- ☐ There is no camber on either the upper or lower surface of the wing.
- ☒ The camber is greater on the upper wing surface than it is on the lower surface of the wing.

10

Which would most likely ensure a safe off-field landing?

- ☒ Maintaining an approach airspeed of at least 50 percent above the glider's stall speed plus half the estimated windspeed.
- ☐ Landing in a pasture or uncultivated field rather than one in cultivation and whose crops have been harvested.
- ☒ Landing into the wind, regardless of the type or slope of the terrain.

11

The best lift/drag ratio of a glider is a value that

- ☒ varies depending upon the weight being carried.
- ☐ remains constant and is independent of the weight being carried.
- ☐ remains constant regardless of airspeed changes.

12

If the nosewheel of an airplane moves aft during gear retraction, how would this aft movement affect the CG location of that airplane? It would

- ☐ cause the CG location to move forward.
- ☐ have no effect on the CG location.
- ☒ cause the CG location to move aft.

13

If the CG of an aircraft is moved from the aft limit to beyond the forward limit, how will it affect the cruising and stalling speed?

- ☐ Increase both the cruising speed and stalling speed.
- ☒ Decrease the cruising speed and increase the stalling speed.
- ☐ Decrease both the cruising speed and stalling speed.

14

A rule of thumb for flying a final approach is to maintain a speed that is

- ☒ 50 percent above the glider's stall speed plus half the estimated windspeed.
- ☐ twice the glider's stall speed plus half the estimated windspeed.
- ☐ 50 percent above the glider's stall speed, regardless of windspeed.

15

When one engine fails on a twin-engine airplane, the resulting performance loss

- ☐ is approximately 50 percent since 50 percent of the normally available thrust is lost.
- ☐ reduces cruise indicated airspeed by 50 percent or more.
- ☒ may reduce the rate of climb by 80 percent or more.

16

What type of stability does the horizontal stabilizer provide during flight?

☐ **Airspeed.**

☒ **Lateral.**

☐ **Longitudinal.**

17

At what bank angle will the resultant of gravity and centrifugal force equal twice a glider's weight?

☒ 60 degree.

☐ 45 degrees.

☐ 30 degrees.

18

The SI units of air density (I) and force (II) are..... &... respectively:

☒ (I) kg / m^3 , (II) N.

☐ (I) N / m^3 , (II) N.

☐ (I) N / kg , (II) kg.

☐ (I) kg / m^2 , (II) kg.

19

Assuming no flow separation and no compressibility effects the location of the centre of pressure of a positively cambered aerofoil section:

- ☐ moves forward when the angle of attack decreases.
- ☐ is at approximately 25% chord irrespective of angle of attack.
- ☐ moves backward when the angle of attack decreases.
- ☒ is independent of angle of attack.

20

The lift formula can be written as:
(rho = density)

☐ $L = n * W.$

☒ $L = CL * \frac{1}{2} \rho * V^2 * S.$

☐ $L = CL * 2\rho * V^2 * S.$

☐ $L = W.$

21

The polar curve of an aerofoil section is a graphic relationship between:

- ☒ lift coefficient (C_L) and drag coefficient (C_D).
- ☐ angle of attack and lift coefficient C_L .
- ☐ drag coefficient C_D and angle of attack.
- ☐ TAS and stall speed.

22

Which of these statements about induced drag are correct or incorrect?

- I. An elliptical spanwise lift distribution generates more induced drag than a rectangular lift distribution.
- II. Induced drag decreases with decreasing aspect ratio.

- ☐ I is correct, II is correct.
- ☒ I is correct, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.

23

An aeroplane transitions from steady straight and level flight into a horizontal co-ordinated turn with a load factor of 2, the speed remains constant and the:

- ☒ total drag increases by a factor of 4.
- ☐ induced drag increases by a factor of 4.
- ☐ lift increases by a factor of 4.
- ☐ angle of attack increases by a factor of 1/4.

24

The induced drag:

- ☒ increases as the lift coefficient increases.
- ☐ has no relation to the lift coefficient.
- ☒ increases as the magnitude of the tip vortices decreases.
- ☐ increases as the aspect ratio increases.

25

Which statement, about the effects on drag of fitting external tip tanks to the wings of an aeroplane, is correct?

- I. Parasite drag decreases.
II. Induced drag increases.

- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is correct, II is incorrect.

26

Which statement, about an aeroplane leaving ground effect at constant angle of attack, is correct?

- I. The lift coefficient C_L remains constant.
II. The induced drag coefficient C_{Di} decreases.

- ☒ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.

27

Which statement, about an aeroplane entering ground effect at constant angle of attack, is correct?

I. The lift coefficient C_L increases.

II. The induced drag coefficient C_{Di} increases.

☐ I is correct, II is correct.

☒ I is incorrect, II is incorrect.

☐ I is correct, II is incorrect.

☐ I is incorrect, II is correct.

28

The stall speed increases, when: (all other factors of importance being constant)

- ☐ weight decreases.
- ☐ spoilers are retracted.
- ☐ minor altitude changes occur e.g. 0-10.000 ft.
- ☒ pulling out of a dive.

29

The effects of very heavy rain (tropical rain) on the aerodynamic characteristics of an aeroplane are:

- ☐ decrease of CLMAX and decrease of drag.
- ☐ increase of CLMAX and decrease of drag.
- ☐ increase of CLMAX and increase of drag.
- ☒ decrease of CLMAX and increase of drag.

30

**Which of the following statements is correct?
Air passes a normal shock wave as**

- ☐ The static temperature increases.
- ☐ The static temperature decreases.
- ☐ The velocity increases.
- ☒ The static pressure decreases.

31

(Please view annex 081-0023)

Assuming no pilot input the motion of the aeroplane in the diagram shows:

- ☒ dynamic longitudinal instability.
- ☐ static longitudinal instability.
- ☐ neutral dynamic longitudinal stability.
- ☐ dynamic longitudinal stability.

See Attachments:

1. 081-0023.gif

Annex

(Attachment
No.2)

32

Which of the following provides a positive contribution to static directional stability?

- ☒ A forward swept wing.
- ☐ A low wing as compared with a high wing.
- ☐ A dorsal fin.
- ☐ Moving the centre of gravity aft.

33

What is the effect on landing speed when a trimmable horizontal stabiliser jams at high IAS?

- ☐ No effect when landing on a high elevation runway.
- ☒ In most cases, no effect.
- ☐ No effect with a forward CG.
- ☐ In most cases, a higher than normal landing speed is required.

34

The most important factor determining the required position of the Trimmable Horizontal Stabiliser (THS) for take off is the:

- ☒ **position of the aeroplane's centre of gravity.**
- ☐ **centre of gravity position of the fuel.**
- ☐ **stall speed.**
- ☐ **total mass of the aeroplane.**

35

Which statement is correct when comparing a fixed pitch propeller with a constant speed propeller?

- I. A constant speed propeller reduces fuel consumption over a range of cruise speeds.
II. A coarse fixed pitch propeller is more efficient during take-off.

☐ I is incorrect, II is incorrect.

☐ I is correct, II is correct.

☒ I is incorrect, II is correct.

☐ I is correct, II is incorrect

36

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch up produces left yaw.
- II. Right yaw produces pitch up.

- ☒ I is correct, II is incorrect.
- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.

37

Which statement is correct regarding the gyroscopic effect of a clockwise rotating propeller on a single engine aeroplane?

- I. Pitch up produces right yaw.
II. Right yaw produces pitch down.

- ☐ I is correct, II is incorrect.
- ☒ I is incorrect, II is incorrect.
- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.

38

Which statement about a propeller is correct?

- I. Asymmetric blade effect is unaffected when engine power is increased.
II. Asymmetric blade effect is independent of the angle between the propeller axis and the
airflow through the propeller disc.

- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.

39

During a straight steady descent, lift is:

- ☒ **equal to weight to maintain a constant speed.**
- ☐ **less than weight,**
- ☐ **greater than weight because it must additionally balance a component of drag.**
- ☐ **less than weight**

40

What factors determine the distance travelled over ground of an aeroplane in a glide from a given altitude?

- ☐ The wind and CLMAX.
- ☐ The wind and the aeroplane's mass.
- ☒ The wind and the lift/drag ratio.
- ☐ The wind and weight together with power loading, which is the ratio of power output to the weight.

Attachment No. 1

Questions 5

figure 147

instrument_147.jpg

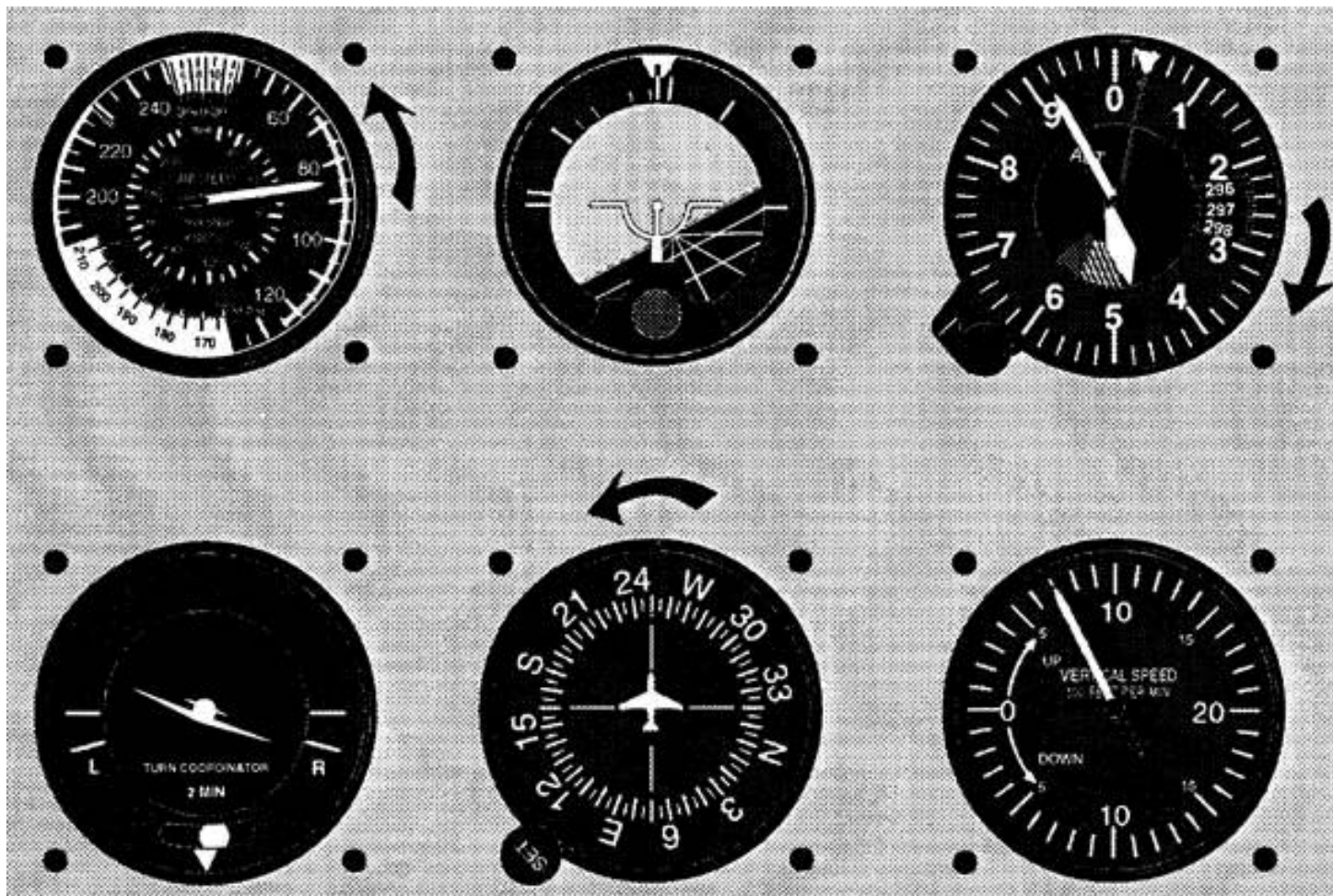


Figure 147. Instrument Sequence (Unusual Attitude) © ASA

Attachment No. 2

Questions 31

Annex

081-0023.gif

081-0023 issue date June 2006

